SentryScan: Assignment 4.1 — Deliberative BDI Modelling & Reconsideration

Course: 21CSE525T — Agent Technology
Assignment: 4.1 — Deliberative BDI Modelling & Reconsideration
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Repository: https://github.com/drgeekg/SentryScan
Target Branch: `assignment/4.1-bdi-modelling`
Git Tag: `assignment-4.1`
Status: Completed & Verified

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1. Executive Summary & Theoretical Context

Assignment 4.1 implements the **Belief–Desire–Intention (BDI) cognitive architecture** (Bratman, 1987; Rao & Georgeff, 1995; Wooldridge, Ch. 4) for SentryScan's Analyst Agent.

1.1 The Classical BDI Paradigm

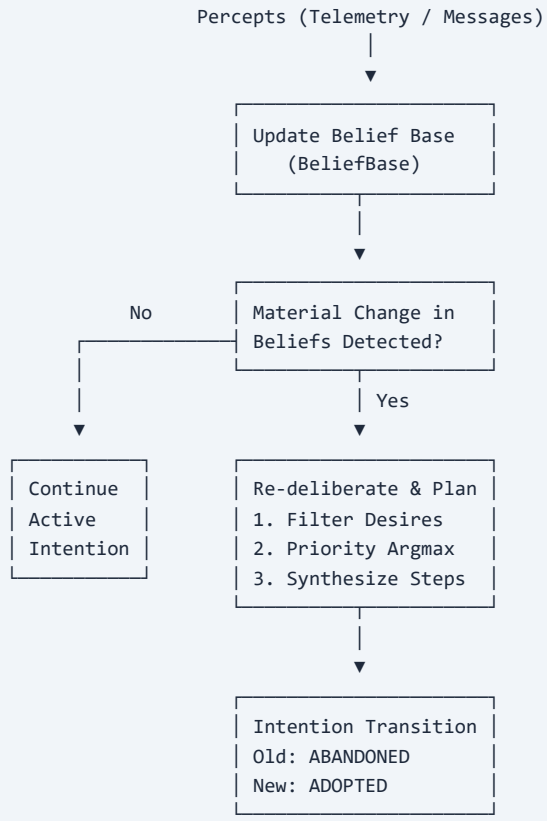
A pure reflex agent reacts instantly without commitment to long-term goals. A BDI agent deliberates and commits to intentions: - **Beliefs (B):** The agent's knowledge about the operational environment (host states, open ports, observed vulnerabilities, active policies). - **Desires (D):** The potential goals or operational states the agent wishes to achieve (audit fleet, contain compromised hosts, surface critical alerts). - **Intentions (I):** The committed desires chosen for execution, organized into concrete procedural plans.

1.2 Intention Reconsideration & Epistemic Preemption

An agent cannot re-deliberate after every minor percept (computational waste), nor can it pursue a plan stubbornly when the environment has changed catastrophically: - **Kinny & Georgeff (1991) Reconsideration Trade-off:** SentryScan implements *bold-agent deliberation* with explicit **material belief update filters**. - **Deliberative Preemption:** Routine telemetry (e.g., informational banner) retains active plans. The arrival of a `CRITICAL` vulnerability finding triggers an immediate interrupt, cleanly abandoning obsolete intentions and adopting emergency containment.

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2. Architecture & Implementation Specification



2.1 Concrete Desires & Intention Synthesis (`src/sentryscan/bdi/deliberation.py`)

Desire Identifier	Eligibility Condition	Intention Plan Synthesized	Action Generated
`CONTAIN_COMPROMISED_HOSTS`	Active `CRITICAL` / `HIGH` finding on unquarantined host	1. `QUARANTINE_HOST` 2. `SET_SCAN_CADENCE (50ms)` 3. `SCHEDULE_PRIORITY_PORTS`	`quarantine_flag=True` `cadence=50ms`
`SURFACE_CRITICAL_FINDINGS`	Host quarantined; findings unacknowledged	1. `PUBLISH_CRITICAL_ALERT` 2. `SET_SCAN_CADENCE (100ms)`	`cadence=100ms`
`AUDIT_ASSIGNED_HOSTS`	Baseline state; no critical alerts active	1. `SET_SCAN_CADENCE (250ms)` 2. `DISPATCH_ROUTINE_PROBE`	`cadence=250ms` (Nominal)

3. Rubric Compliance & Verification Matrix

Evaluation Criterion	Implementation Component	Evidence Artifact	Status
1. Distinct Beliefs, Desires, and Intentions	`src/sentryscan/bdi/beliefs.py`, `src/sentryscan/bdi/desires.py`, `src/sentryscan/bdi/intentions.py`: Epistemic separation into `BeliefBase`, `Desire`, and `Intention`	`tests/unit/test_bdi_models.py` `evidence/bdi_transition_log.md`	**PASSED**
2. Deliberation & Option Selection	`src/sentryscan/bdi/deliberation.py`: Option generation, feasibility filtering, and dynamic priority ranking selecting $d^* = \arg\max Priority(d)$	`tests/unit/test_bdi_deliberation.py` `demos/demo_assignment_4_1.py` Cycle 1	**PASSED**
3. Plan Synthesis & Intention Steps	`src/sentryscan/bdi/deliberation.py`: Instantiates concrete ordered steps (`QUARANTINE_HOST`, `SET_SCAN_CADENCE`, etc.) with status tracking	`tests/unit/test_bdi_deliberation.py` `evidence/bdi_transition_log.md`	**PASSED**
4. LangGraph Deliberative Workflow	`src/sentryscan/graphs/analyst_graph.py`: Compiled LangGraph orchestrating `perceive` $ o$ `check_reconsideration` $ o$ `deliberate_and_plan` $ o$ `execute_step`	`tests/unit/test_analyst_graph.py` `src/sentryscan/agents/analyst.py`	**PASSED**
5. Material Reconsideration Trigger	`src/sentryscan/graphs/analyst_graph.py`: Detects escalations to `CRITICAL` / `HIGH` severity and triggers intentional preemption	`tests/integration/test_bdi_reconsideration.py` `demos/demo_assignment_4_1.py` Cycle 3	**PASSED**
6. Clean Intention Abandonment & Transition	Previous intention `intention-000001` cleanly transitioned to `ABANDONED` with recorded rationale rather than silent overwriting	`tests/integration/test_bdi_reconsideration.py` `evidence/bdi_transition_log.md` Cycle 3	**PASSED**
7. Emergency Containment Execution	Autonomous quarantine flag (`quarantine_flag=True`) and high-frequency sampling (`50ms`) emitted via `PARAMS_UPDATE` message	`tests/unit/test_analyst.py` `demos/demo_assignment_4_1.py` Cycle 3	**PASSED**
8. Post-Containment Intention Shift	Following host quarantine, agent autonomously transitions from `CONTAIN_COMPROMISED_HOSTS` to `SURFACE_CRITICAL_FINDINGS`	`evidence/bdi_transition_log.md` Cycle 4	**PASSED**
9. Trace & Single-Writer Auditing	Sequential event log recording message reception, state transitions, and generated parameter adaptations via `src/sentryscan/observability/recorder.py`	`evidence/trace_excerpt.jsonl`	**PASSED**

4. Execution Results & Empirical Evidence

4.1 BDI State Transition & Reconsideration Log (`evidence/bdi_transition_log.md`)

Cycle	Trigger	Target Host	Severity	Old Intention	New Intention	Reason	Generated Action
1 (Init)	`INITIAL_DELIBERATION`	`192.0.2.10`	`INFO`	None	`intention-000001` (AUDIT)	Initial boot	`NOMINAL (250ms)`
2 (Routine)	`ROUTINE_TELEMETRY`	`192.0.2.20`	`LOW`	`intention-000001`	Retained (No Change)	Non-material	`NOMINAL (500ms)`
3 (Preempt)	`MATERIAL_BELIEF_UPDATE`	`sim-host-007`	**`CRITICAL`**	`intention-000001` [ABANDONED]	`intention-000002` (CONTAIN)	Critical alert on host	`QUARANTINE=True (50ms)`
4 (Shift)	`CONTAINMENT_SATISFIED`	`sim-host-007`	`CRITICAL (Quar)`	`intention-000002` [SUCCEEDED]	`intention-000003` (SURFACE)	Containment satisfied	`DEEP_INSPECT (100ms)`

5. Submission Package & Reproduction Instructions

5.1 Exact Submission Components

1. **Report Document:** `assignments/assignment_4_1/deliverables/SentryScan_Assignment_4_1_Report.pdf`
2. **Repository URL:** `https://github.com/drgeekg/SentryScan`
3. **Target Branch:** `assignment/4.1-bdi-modelling`
4. **Immutable Git Tag:** `assignment-4.1`
5. **Packaged ZIP Archive:** `dist/SentryScan_assignment_4_1.zip`
6. **SHA-256 Checksum:** `dist/SentryScan_assignment_4_1.zip.sha256`

5.2 Verification Commands

```
# Run BDI deliberation demonstration
python demos/demo_assignment_4_1.py

# Execute BDI unit and integration tests
pytest -v tests/unit/test_bdi_models.py tests/unit/test_bdi_deliberation.py tests/integration/test_bdi_reconsideration.py
```